American Computer Science League

2024 Finals ● Short Problems ● Senior Division

1. Boolean Algebra

How many ordered triples make the following Boolean expression TRUE?

$$\overline{A(B + \overline{C})} \oplus \overline{ABC}$$

A. 0
B. 2
C. 4
D. 6
E. None of the above

2. Boolean Algebra

Simplify the following Boolean expression:

$$A \oplus (\overline{B\,C} + \overline{A\,B\,C})$$

A. $\overline{A} + A\overline{B}C$
B. $\overline{A} + ABC$
C. $A + ABC$
D. $\overline{A} + AB\overline{C}$
E. None of the above

3. Bit-String Flicking

Evaluate the following expression:

(RCIRC-5 11011010) XOR (LSHIFT-3 10101101)
AND 11001101 OR 10001010

A. 10011110
B. 10101110
C. 10110110
D. 10010100
E. None of the above

4. Bit-String Flicking

How many values of X, a bit-string of 5 bits, make the following equation TRUE?

(RCIRC-1 (LSHIFT-2 X)) OR (LCIRC-3 (RSHIFT-2 X)) = X

A. 1
B. 2
C. 3
D. 4
E. None of the above

5. Recursive Functions

Find $f(15, 11)$, given:

$$f(x,y) = \begin{cases} 2 \cdot f(x-2,\ y-3) + 1 & \text{if } x \geq 10 \\ f(y-1,\ x-3) - 1 & \text{if } 4 < x < 10 \\ x \cdot y & \text{if } x \leq 4 \end{cases}$$

A. 5
B. 6
C. 11
D. 23
E. None of the above

6. Recursive Functions

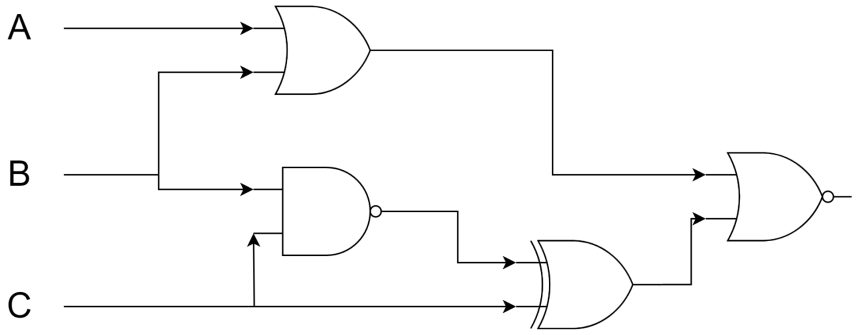
Find $f(f(20))$, given:

$$f(x) = \begin{cases} f(x-4) - 3 & If\ x \geq 12 \\ f(x-3) + 2 & if\ 6 < x < 12 \\ x^2 & if\ x \leq 6 \end{cases}$$

A. 14
B. 16
C. 18
D. 25
E. None of the above

7. Digital Electronics

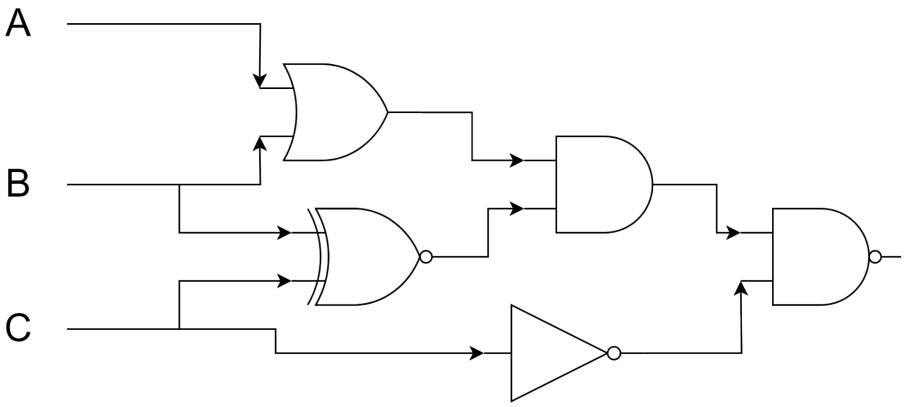
Which Boolean expression is equivalent to the following digital circuit?



A. $A\,B\,C$
B. $A\,\overline{B}\,C$
C. $\overline{A}\,B\,C$
D. $\overline{A}\,\overline{B}\,C$
E. None of the above

8. Digital Electronics

How many ordered triples make the following digital circuit FALSE?



A. 1
B. 2
C. 3
D. 4
E. None of the above

9. Prefix-Infix-Postfix

Define the ternary operator # as the sum of its 3 operands divided by 2.
Define the unary operator \/ as the principal square root of its operand.
Find the value of A using the following postfix equations:

s 5 3 4 * 6 7 + # =
A s s 5 – * s 6 2 * – s 5 2 * 3 + – * * \/ =

A. 12
B. 15
C. 24
D. 30
E. None of the above

10. Prefix-Infix-Postfix

Convert the following postfix expression to prefix if all numbers are single digits:

2 4 8 2 − * 5 2 ^ / − 8 4 / 2 / 8 * +

A. +−2*/4−82^52*//8428
B. +−2/*4−82^52*//8428
C. +−2/*482−^52*//8428
D. +−2/*4−82^52*//8248
E. None of the above

11. Computer Number Systems

What is the value of the following expression in hexadecimal?

A_{16} * (42_8 − F_{16} + 25_{10}) / 16_{16}

A. 14
B. 16
C. 18
D. 20
E. None of the above

12. Computer Number Systems

What is the final octal digit of 3_8 ^ 11_{10}?

A. 1
B. 3
C. 5
D. 7
E. None of the above

13. Data Structures

Build a binary search tree for:

REFRIGERATOR

What is the postorder trasversal for this tree?

A. ROGRIAREFETR
B. AEGOIFERRTRR
C. AEGRORIRFETR
D. AEGRORRIFETR
E. None of the above

14. Data Structures

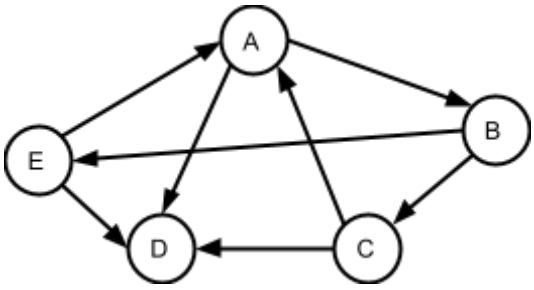
The following operations performed on a stack can be used to evaluate a postfix expression. If the result is Z, what is the postfix expression that would be evaluated by these stack operations?

```
PUSH(5)
PUSH(7)
X = POP()
Y = POP()
PUSH(Y*X)
PUSH(9)
PUSH(3)
X = POP()
Y = POP()
PUSH(Y/X)
PUSH(2)
X = POP()
Y = POP()
PUSH(Y+X)
X = POP()
Y = POP()
PUSH (Y/X)
PUSH(2)
X = POP()
Y = POP()
PUSH(Y-X)
Z = POP()
```

A. 5 7 * 9 3 2 / + / 2 −
B. 5 7 * 9 3 / 2 + / 2 −
C. 5 7 * 9 3 / 2 2 / + −
D. 5 7 * 9 3 / 2 2 / − +
E. None of the above

15. Graph Theory

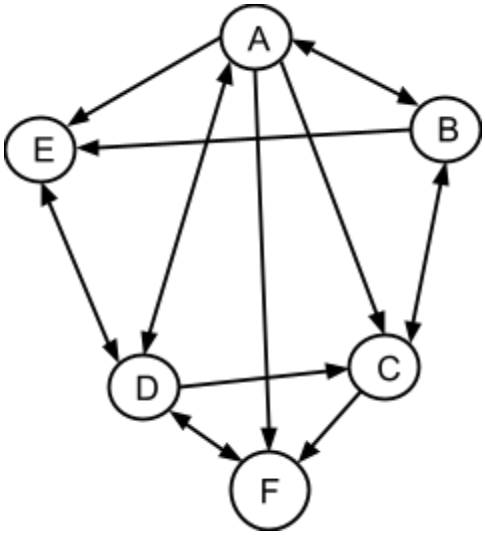
Given the directed graph below, what is the total number of paths of length 3?



A. 9
B. 10
C. 11
D. 12
E. None of the above

16. Graph Theory

How many cycles are there from C in the directed graph below?



A. 15
B. 10
C. 6
D. 3
E. None of the above

17. What Does This Program Do?

What is the output when the following program is executed?

```
for p = 2 to 4
    val = 2^p * (2^(p+1)-1)
    sum1 = 0
    for y = 1 to 2^(p+1)-1
        sum1 = sum1 + y
    next y
    sum2 = 0
    for y = 1 to 2*p-1 step 2
        sum2 = sum2 + y^3
    next y
    if val == sum1 && val == sum2 then
        output p
    end if
next p
```

A. 2, 3, 4
B. 2, 4
C. 3, 4
D. 2, 3
E. only 1 of them

18. LISP

Evaluate the following sequence of LISP expressions:

```
(SETQ G '(c (a) (a c)))
(SETQ H '((a b (c a)) (a (b))))
(CDADR (REVERSE (CONS G (CDDAR H))))
```

A. (c)
B. (a a c)
C. ((a c))
D. ((a) (a c))
E. None of the above

19. Regular Expressions

How many of the following strings are accepted by the regular expression:

```
[a-lo-r][aeiou]s?[^u-v]*
```

a) fortran b) cobol c) java
d) lisp e) assembly f) machine
g) python h) basic i) pascal

A. 1
B. 3
C. 5
D. 7
E. None of the above

20. Assembly Language

What is the output when the following program is executed?

```
N       DC      1
T       DC      1
M       DC      100
        LOAD    T
TOP     SUB     M
        BG      STOP
        LOAD    N
        ADD     =1
        STORE   N
        LOAD    T
        ADD     N
        STORE   T
        BU      TOP
STOP    LOAD    T
        SUB     N
        STORE   T
        PRINT   T
        END
```

A. 6
B. 28
C. 91
D. 105
E. None of the above